Miljkovic and Wang *Financial Innovation* (2025) 11:29
https://doi.org/10.1186/s40854-024-00689-1

Financial Innovation

RESEARCH Open Access

# A dimension reduction assisted credit scoring method for big data with categorical features

Tatjana Miljkovic[1*] and Pei Wang[2]

*Correspondence: miljkot@miamioh.edu

[1] Miami University, Department of Statistics, 100 Bishop Circle, 334E Upham Hall, Oxford, OH 45056, USA
[2] Department of Applied Statistics and Operations Research, Bowling Green State University, 819 E Wooster St, Bowling Green, OH 43403, USA

**Abstract**

In the past decade, financial institutions have invested significant efforts in the development of accurate analytical credit scoring models. The evidence suggests that even small improvements in the accuracy of existing credit-scoring models may optimize profits while effectively managing risk exposure. Despite continuing efforts, the majority of existing credit scoring models still include some judgment-based assumptions that are sometimes supported by the significant findings of previous studies but are not validated using the institution's internal data. We argue that current studies related to the development of credit scoring models have largely ignored recent developments in statistical methods for sufficient dimension reduction. To contribute to the field of financial innovation, this study proposes a Dimension Reduction Assisted Credit Scoring (DRA-CS) method via distance covariance-based sufficient dimension reduction (DCOV-SDR) in Majorization-Minimization (MM) algorithm. First, in the presence of a large number of variables, the DRA-CS method results in greater dimension reduction and better prediction accuracy than the other methods used for dimension reduction. Second, when the DRA-CS method is employed with logistic regression, it outperforms existing methods based on different variable selection techniques. This study argues that the DRA-CS method should be used by financial institutions as a financial innovation tool to analyze high-dimensional customer datasets and improve the accuracy of existing credit scoring methods.

**Keywords:** Credit scoring, Dimension reduction, Logistic regression, Majorization-minimization algorithm

**Mathematics Subject Classification:** C40, D91, E41, G21, G41

## Introduction

### Background

Over the past 30 years, financial institutions have relied on credit-scoring models to assess consumers' likelihood of paying a loan on time. Credit scoring models may classify the applicants into "low," "medium," and "high" risk classes. An applicant with a high credit score and steady income is likely to be perceived as having low" risk and will therefore receive a low interest rate on the loan. In contrast, an applicant with a poor credit

Springer Open

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

history is likely to be perceived as " high-risk and may be required to obtain financing from a bank (or lender) that offers loans with relatively high interest rates. Based on risk classification, financial institutions decide whether to lend financial resources to applicants with agreed-upon terms and conditions. In the financial sector, applicants are not only individuals but can also include small and medium-sized enterprises (SMEs), which have an important influence on economic development (Roy and Shaw 2021).

Today, as major commercial banks hold billions of dollars in consumer loans, the consumer credit sector is creating a huge industry of significant economic importance. Thus, the accuracy of credit-scoring models has become a major focus for financial institutions to optimize profits while effectively managing risk exposure. Small improvements in the accuracy of credit-scoring models may result in considerable financial gains and reduce the misclassification rates for consumers (Gunnarsson et al. 2021).

Markov et al. (2022) state that credit risk is subject to external evaluation as financial institutions are constantly monitored by central banks and auditors to make sure they comply with Basel and International Financial reporting Standards (IFRS). Following the recommendations of Basel II (BCBS, 2006), financial institutions are advised to use their own datasets that are representative of their diverse customer domains to ensure that credit scoring methods can be evaluated under a wide range of conditions. A credit-scoring model that serves one commercial bank well may not be a suitable benchmark tool for another commercial bank with a significantly different consumer profile. Additionally, a phenomenon known as population drift occurs because customer populations tend to evolve over time, causing changes in distributions (Hand and Henley 1997). This can be a problem in credit scoring applications where particular variable screening practices have been used for a long time without considering innovations in this area of research. The applicant population is subject to demographic changes, economic pressures, and competitive environments. If these factors are not considered, various sources of bias may be introduced, causing significant concerns for financial institutions and customers.

### What is lacking?

Despite continuing efforts, the majority of existing credit scoring models still rely on judgment-based assumptions and expert knowledge and ignore many informative features of the data. It is essential to recognize that the rapid increase in consumer data size and volume has created a basis for innovation, differentiation, and growth. To compete in this new data-driven era, financial institutions must leverage big data to develop credit scoring models. We argue that current studies related to the development of credit-scoring models omit recent developments in statistical methods for sufficient dimension reduction.

Dimension reduction is an important statistical technique that minimizes the number of variables in a data set without losing valuable information. Historically, credit-scoring models have employed dimension-reduction methods to select a subset of variables necessary to build good predictive models. Tufféry (2011) summarized these methods in the context of building credit scoring models: partial least squares, stepwise regression, forward selection, backward elimination, lasso, variable elimination due to multicollinearity, variable elimination due to a large percentage of missing values, and

principal component analysis (PCA). PCA, originally invented by Pearson (1901), is the most popular linear dimension reduction approach. With the increasing use of PCA in a wide range of applications, Uyeda et al. (2015) discussed statistical and conceptual issues that may lead to misleading inferences. For example, the performance of PCA may be affected by correlations in the data.

Khandani et al. (2010) noted that a massive amount of transaction data combined with individual consumer data and account balance data creates significant challenges, even in developing basic intuition regarding the data. The authors considered a large transaction dataset consisting of 138 categories provided by a major US consumer bank. However, only a small subset of the data was used to predict credit card defaults and delinquencies. Expert knowledge, experience, and references to existing studies were used to reduce a large number of data features to a smaller subset for model building. A recent study by Woo and Sohn (2022) considers the Landing Club dataset, which includes 145 initial variables with borrower information. However, the authors used only 13 variables (seven continuous original variables, two continuous derived variables, and four categorical variables) with high importance, based on the findings of previous studies (Emekter et al. 2015; Guiso et al. 2013). The authors claimed that the other variables were removed owing to noise, redundancy, and data complexity, which could decrease the risk of overfitting and mitigate the curse of dimensionality. In addition, redundant features may not have contributed to the predictive accuracy of the model. Given the sensitivity of consumer data, many studies on credit-scoring methods have lacked transparency and reproducibility.

#### How do we add value?

Although much research has been conducted, the credit scoring literature does not reflect several recent advancements in the area of Sufficient Dimension Reduction (SDR) based on distance covariance-based techniques (e.g., (Sheng and Yin 2013, 2016), Wu and Chen (2021)). While various dimension-reduction methods have been explored in previous credit-scoring studies, an optimal dimension-reduction technique for building a reliable predictive credit-scoring model has not been sufficiently studied and remains a potential area of research. To add value to the field of financial innovation, this study proposes the DRA-CS method based on the distance covariance-based sufficient dimension reduction (DCOV-SDR) technique by considering the following research questions:

- Q1. What variable features contribute the most in explaining creditworthiness for an individual consumer?
- Q2. How is the credit score calculated based on the proposed DRA-CS method?
- Q3. What are the advantages of our proposed DRA-CS method compared to the PCA method?

Our work is motivated by Wu and Chen (2021)'s previously published study on using Riemannian Newton's optimization algorithm with the majorization-minimization method (MMRN) to solve the DCOV-SDR. The MMRN algorithm is computationally efficient and robust across various data settings compared to other algorithms for conducting DCOV-SDR, for example, the sequential quadratic programming (SQP) used in

Sheng and Yin (2013, 2016). According to the simulation studies in Wu and Chen (2021), the MMRN algorithm can be 25 times faster than the SQP algorithm and is more robust because the standard deviation of the computational time is smaller. In a real-data application, Wu and Chen (2021) observed that the MMRN algorithm was 47 times faster than SQP. Extensive computational comparisons are presented in the study published by Wu and Chen (2021). Thus, it is suitable for credit-scoring applications that include many variables.

Transaction data collected by banks tend to be characterized by a large number of categorical predictors with multiple levels. The DRA-CS approach is well-suited for handling these types of datasets. Financial institutions can use our model to estimate the credit scores of new applicants for credit, loans, or insurance products. Sensitivity analysis applied to the DRA-CS model on artificial datasets showed promising results for different settings, indicating that our model outperformed the classical PCA approach.

To enhance the reproducibility of our research, we employed a publicly available high-dimensional dataset, particularly when applying dummy coding for categorical variables in the three-step credit-scoring model, which encompasses 59 variables. The exploration of dimension reduction techniques for similarly sized high-dimensional data is common practice (see Wang and Yin (2008), (Sheng and Yin 2013, 2016), Wu and Chen (2021), Dumitrescu et al. (2022), Wang et al. (2021) Wang and Xue (2023), among others). Furthermore, two other publicly available datasets were included to assess the robustness of the results, allowing for full reproducibility and transparency of our findings. This approach aims to promote an exchange of ideas and contribute to the collective advancement of research in global academic forums.

The remainder of this paper is organized as follows. It begins with a literature review of variable selection techniques and current regression models in credit scoring. Subsequently, the methodology and dataset are outlined, followed by an analysis of the results and a discussion. Then, a sensitivity analysis through a simulation study and robustness testing of the results to the choice of a dataset are explored. Finally, the study concludes with a discussion of the results.

## Literature review

The growth of big data has attracted considerable attention from the statistics and machine learning research communities. Consequently, various methods have been proposed to reduce the dimensionality of data with vast dimensions without losing important information. Dimension reduction is highly important for financial institutions that process large amounts of data on a daily basis, including managing, interpreting, and analyzing the data to develop empirical models to support credit-based decisions. An increase in the availability of customers' social and psychometric information creates an opportunity for financial institutions to consider the psychometric properties of borrowers, such as personality and sentiments, to build credit scoring models. Thus, an efficient dimension reduction approach is highly important in credit scoring.

Table 1 shows a chronological summary of numerous previous studies that incorporated various dimension reduction methods into credit scoring models. An earlier study by Hand and Henley (1997) presented three approaches for reducing variables: expert knowledge, stepwise procedures, and the measure of information value. In practice,

**Table 1** Overview of related literature analyzing dimension reduction methods with classification methods for credit scoring

| Authors | Data sets | Features | Scoring methods | Dimension reduction | Main findings/ recommendations |
|---|---|---|---|---|---|
| Hand and Henley (1997) (review article) | N/A | N/A | LOR, DA, MP, DT, ES, NN, NNM, TVM | Expert knowledge, stepwise, information value | There is no overall best method; data structure and characteristics are relevant NN and NNM are sensitive to overfitting |
| Liu and Schumann (2005) | 1 | 72<br>selection varies by method | MT, NN, LOR, *k*-NN | REF, CFS, CON, WRP | CON and WRP perform well in reducing the number of features |
| Khandani et al. (2010) | 1 | 138<br><40 selected | CART | N/A | Estimated cost savings from 6% to 25% of total losses |
| Tufféry (2011) | 1 | 20<br>8 selected | LR | stepwise, forward, backward | LOR remains one of the most reliable classification methods |
| Wang et al. (2012) | 2 | 14<br>15 | NN, DT, LOR | RSFS | RSFS is less expensive in computational cost. |
| Sadatrasoul et al. (2015) | 2 | 24<br>14 | Hybrid Fuzzy Apriori | stepwise CART, PCA Pearson correlation | PCA is the best, CART is placed the second in the number of rule's measure. Stepwise outperforms CART in the feature extraction measure. |
| Trivedi (2020) | 1 | 20<br>15 selected | B, NB, DT, SVM, RF | Chi-Square, IG, GR | Chi-Square with RF or DT show good performance |
| Giannouli et al. (2021) | 2 | 39 (small business)<br>49 (large business)<br>selection varies by method | LOR | stepAIC PCA, ENR | StepAIC, resulted in models containing 38% (small business) and 36%(large business) of the variables used in the full model. |
| Woo and Sohn (2022) | 1 | 145<br>< 20 selected | Locally weighted LOR | based on existing literature | Study provides insights into alternative credit scoring for P2P lending |
| Dumitrescu et al. (2022) | 4 | 10<br>12<br>14<br>23 | PLTR | Adaptive Lasso decision trees | Predicts credit risks significantly more accurately than LOR |
| Deng and Wang (2023) | 2 | 32<br>23 | *k*-NN, XGBoost penalized LOR | SFS Lasso | XGBoost performs well |

these approaches are used in a sequential and iterative manner with statistical scoring models, such as Logistic Regression (LOR), Discriminant Analysis (DA), Mathematical Programming (MP), Decision Trees (DT), expert systems (ES), Neural Networks (NN), the Nearest-Neighbor Method (NNM), and time-varying models (TVM). Although Hand and Henley (1997) provided an excellent review of statistical classification

methods, it was clear that the process of selecting the best features was somewhat unsystematic and arbitrary. To resolve this issue, Liu and Schumann (2005) proposed four machine-learning methods for feature selection by reducing the feature space. These methods include ReliefF (REF), correlation-based features (CFS), consistency-based features (CON), and wrapper feature (WRP) algorithms. An analysis of real data with 72 features and four classification algorithms (model tree (MT), NN, LOR, and k-nearest neighbor (*k*-NN)) showed that CON and WRP performed best in terms of error rate for reducing the number of features. In the study conducted by Khandani et al. (2010), transaction data with 138 features were considered, but only a small subset of features was used to build a generalized classification and regression tree (CART), a machine-learning model of consumer credit risk. Owing to legal restrictions and data confidentiality, additional information on the method used for feature selection was not provided; therefore, it remains unclear. An excellent book Tufféry (2011) explored various methods to integrate LOR, DA, and Principal Component Analysis (PCA) in the computation of credit scores. The author highlighted that most standard statistical software include built-in stepwise selection procedures (i.e., forward, backward, and in-step selection) in the regression framework, which makes these models attractive to users. It should be noted that LOR remains one of the most reliable classification models.

Another popular method used in dimension reduction is Lasso. Lasso is a regularization technique proposed by Tibshirani (1996) and is employed for simultaneous estimation and variable selection in the regression framework. Zou (2006) proposed improvements to Lasso by introducing Adaptive Lasso, addressing some of its limitations, especially when dealing with correlated predictors. The primary idea behind Adaptive Lasso is to assign different penalty weights to different coefficients based on their estimated importance or relevance. To overcome certain limitations of the LASSO, Zou and Hastie (2005) introduced an Elastic Net as a regularization technique. Similar to the Lasso, the Elastic Net performs automatic variable selection and continuous shrinkage, allowing the selection of groups of correlated variables. It can be compared to a stretchable fishing net that retains "all the big fish." The authors demonstrated that the Elastic Net often outperforms the lasso in terms of prediction accuracy. These methods have gained popularity in statistics and machine learning, and have found many applications.

In the field of credit scoring, researchers, including Ogundimu (2022), have explored the application of the Lasso and Adaptive Lasso. More recently, Dumitrescu et al. (2022) introduced the penalized logistic tree regression (PLTR) method, which leverages information from decision trees to enhance the predictive power of logistic regression. The PLTR uses a logistic regression model with linear predictors extracted from decision trees. To handle a possibly large number of decision-tree rules, the authors incorporated an Adaptive Lasso as a variable selection technique. The results showed that the PLTR model not only increased the accuracy of logistic regression but also significantly reduced misclassification costs (see Viaene and Dedene (2005)).

Feature selection techniques using machine-learning approaches were studied by Trivedi (2020). The author considered three methods for feature selection (Chi-Square,

Information Gain (IG), and Gain Ratio (GR)) and five Machine Learning Classifiers (Bayesian (B), Naive Bayes (NB), Support Vector Machines (SVM), Decision Trees (DT), and Random Forest (RF)). The RF method was found to be the best, reporting 93% accuracy and a reduction in False Positive (Type 1 error) and False Negative (Type 2 error) rates in the credit scoring application. Laborda and Ryoo (2021) used one filter method (Chi-squared test and correlation coefficients) and two wrapper methods (forward stepwise selection and backward stepwise selection) to mitigate the overfitting in the curse of dimensionality. The results suggest that forward stepwise selection yields superior performance for each classification method, including logistic regression, support vector machines, K-nearest neighbors, and random forest. A study by Deng and Wang (2023) compared penalized logistic regression, Extreme Gradient Boosting (XGBoost), and *k*-nearest neighbor with sequential forward selection on two different credit datasets. The findings revealed that XGBoost outperformed the other methods when employed in dimension-reduction techniques.

Giannouli et al. (2021) considered stepAIC, PCA, and optional Elastic Net Regularization (ENR) as the dimension reduction techniques employed with LOR for developing credit models for small and large enterprises registered through the Credit Bureau operating in Greece. Dimension reduction was achieved at two levels: first, by applying the stepAIC procedure to the standardized variables, and then by performing PCA on the variables selected by stepAIC. ENR is recommended if further dimensionality reduction is required.

Woo and Sohn (2022) considered the Lending Club dataset with 145 variables in building the credit scoring model in peer-to-peer (P2P) landing. Although many psychometric features are available in the data, only a small set of variables whose coefficients have been reported to be important or significant in the literature were selected. One needs to be cautious with this approach in selecting features because it may introduce a large bias into the results owing to a significant loss of information. Deng and Wang (2023) used the sequential forward selection (SFS) method to perform variable selection by adding variables to the model one by one. The *k*-NN approach is used to stop the selection process when the accuracy of the model ceases to increase.

The credit scoring literature does not reflect the recent advancements in SDR. In regression analysis, the SDR provides a useful statistical framework for analyzing high-dimensional datasets without losing information (Wu and Chen 2021). Many SDR methods are available in previous studies, such as sliced inverse regression (SIR; Li (1991)) and sliced average variance estimation (SAVE; Cook and Weisberg (1991)). While they are promising for reducing data size, many require strong conditions (e.g., certain model assumptions or a normal distribution for the predictor). These conditions were relaxed by introducing the distance covariance (DCOV) measure, which quantifies the dependence strength between two random vectors (Székely et al. 2007), further leading to robust SDR methods (model-free and requiring very mild conditions). Thus, DCOV-based SDR methods that focus on solving complex optimization problems have been introduced (Sheng and Yin 2013, 2016; Chen et al. 2018). A complex optimization problem for solving the SDR was also explored by Wu and Chen (2021). Although these SDR

methods have been applied in various statistical contexts, they have not been considered for credit scoring applications. Our study extends the existing research on feature selection and dimension reduction by introducing a novel SDR method for credit scoring. The choice to utilize the distance-covariance-based SDR stems from the specific characteristics of the data, particularly the presence of categorical variables. However, one needs to be cautious not to compare dimension-reduction approaches directly with the current state-of-the-art credit scoring methods, as this comparison may lead to unfair assessments owing to distinct data utilization mechanisms.

## Data and methodology

### Data description

This study uses the dataset `credit`, which contains credit information for 1,000 German consumers, of which 300 are classified as defaulting. These data are available in the R (R Core Team (2022)) package **CAS datasets** developed by Dutang and Charpentier (2016). There are 21 explanatory variables of interest in our analysis, which are summarized in Table 2. Fifteen variables were categorical with multiple levels, whereas six were numerical. This dataset has been used by other authors (i.e., Tufféry (2011)) to apply data

**Table 2** Summary of the variables included in the `credit` data set

| **Categorical variables** | |
|---|---|
| **Variable name** | **Short description with categorical levels** |
| Checking status | Status of existing checking account (4 levels) |
| Credit history | Credit history (5 levels) |
| Purpose | Purpose of credit (10 levels) |
| Savings | Savings amount (5 levels) |
| Employment | Present employment status (5 levels) |
| Personal status | Personal status and sex (4 levels) |
| Other parties | Other debtors or guarantors (3 levels) |
| Residence since | Present residence since (4 levels) |
| Property magnitude | Property (most valuable) (4 levels) |
| Other payment plans | Other installment plans (3 levels) |
| Housing | Housing type (3 levels) |
| Job | Level of skills required in job (4 levels) |
| Telephone | Whether the telephone is registered under customer name or not (2 levels) |
| Foreign worker | Whether a worker has foreign citizenship or not (2 levels) |
| Class | Credit-worthy against not credit-worthy (2 levels) |
| **Numerical Variables** | |
| **Variable Name** | **Short description** |
| Duration | Credit duration in months |
| Credit amount | Credit amount in Deutsche marks |
| Installment rate | Installment rate (in percentage of disposable income) |
| Age | Age (above 18 years) |
| Existing credits | Number of existing credits at this bank (including the running one) |
| Number of dependents | Number of people being liable to provide maintenance |

mining techniques to compute credit scores that the banking and insurance industries can use.

The `credit` dataset is integrated in this study to illustrate two applications: 1) demonstrate how the MMRN-based DCOV-SDR (MD-SDR) method can be applied to dimension reduction and 2) build a predictive model that will be used to compute the credit score for a new credit or loan applicant. This study focuses on the largest number of categorical variables at multiple levels, which are often seen in account-level transaction data for a customer base. The R functions employed in this study are fully available upon request.

Variables such as duration, amount, and purpose are related to credit itself. Other variables such as outstanding amounts or borrowings relate to the borrower's financial profile. Finally, borrowers' personal information is provided through variables such as age and marital status.

Following Tufféry (2011), the numerical variables are further classified into classes by applying the discretization method, which enables them to be used together with other variables for simplicity and improved readability. For example, the variable age was discretized into two classes: ages less than or equal to 25 and ages greater than 25. See Table 7 for detailed discretization. The variable class is used as the response variable when building the predictive model.

#### MMRN-based DCOV-SDR

The MM algorithm is one of the most efficient optimization algorithms for finding the maxima or minima of a concave/convex objective function with a low computational cost that can handle big data (Ortega and Rheinboldt 2000). The MM algorithm involves two steps: developing a surrogate function to locally approximate the objective function with minimum difference, and then maximizing the surrogate function (see details in Sun et al. (2016)). Wu and Chen (2021) applied the MM algorithm to the DCOV-based SDR problem whose objective function is given by

$$\mathscr{V}_n^2(\boldsymbol{\beta}^T\mathbf{X},\mathbf{Y}) := \frac{1}{n^2}\sum_{k,l=1}^{n} a_{kl}(\boldsymbol{\beta})\boldsymbol{B}_{kl}, \quad \text{s.t.} \quad \boldsymbol{\beta}^T\hat{\boldsymbol{\Sigma}}\boldsymbol{\beta} = I_d \tag{0.1}$$

where $\mathbf{X}$ and $\mathbf{Y}$ denote the predictor and response variables respectively, $\boldsymbol{\beta}$ is the target basis, $\hat{\boldsymbol{\Sigma}}$ is the sample covariance matrix of $\mathbf{X}$, $I_d$ is a $d \times d$ identity matrix, $a_{kl}(\boldsymbol{\beta}) = \| \boldsymbol{\beta}^T\boldsymbol{X}_k - \boldsymbol{\beta}^T\boldsymbol{X}_l \|_2$, and $\| \cdot \|_2$ denotes the Euclidean norm. Here $\boldsymbol{B}_{kl} = b_{kl} - \bar{b}_{k\cdot} - \bar{b}_{\cdot l} + \bar{b}_{\cdot\cdot}$ where $b_{kl} = \| \boldsymbol{Y}_k - \boldsymbol{Y}_l \|_2$, $\bar{b}_{k\cdot} = \frac{1}{n}\sum_{l=1}^{n} b_{kl}$, $\bar{b}_{\cdot l} = \frac{1}{n}\sum_{k=1}^{n} b_{kl}$, and $\bar{b}_{\cdot\cdot} = \frac{1}{n^2}\sum_{k,l=1}^{n} b_{kl}$.

The non-differentiability of the objective function allowed Wu and Chen (2021) to add a perturbation term to $a_{kl}(\boldsymbol{\beta})$ and construct surrogate functions by supporting the hyperplane inequality. To maximize the surrogate function, a Newton's step is used over the Stiefel manifold. This involves solving a direction (Newton's direction) from the

Riemannian Newton's equation and then updating the target basis $\boldsymbol{\beta}$ through the Newton's direction. Algorithm 1 summarises these steps.

**Algorithm 1** MD-SDR Algorithm

**Data:** Input $\mathbf{X}$, $\mathbf{Y}$, and a pertubation constant $\epsilon > 0$.
**Input:** Compute $\hat{\boldsymbol{\Sigma}}$, initialize a $p \times d$ matrix $\boldsymbol{\beta}^{(0)}$ s.t. $\boldsymbol{\beta}^{(0)^T}\hat{\boldsymbol{\Sigma}}\boldsymbol{\beta}^{(0)} = I_d$.
**Step 0:** Calculate $\boldsymbol{B}_{kl}$.
**Step 1:** Update the surrogate function.
**Step 2:** Solve a direction from Riemannian Newton's equation.
**Step 3:** Update the estimated basis $\hat{\boldsymbol{\beta}}$
**Repeat:** Steps 1-3 until the objective function converges.
**Result:** Output $\hat{\boldsymbol{\beta}}$.

Based on the MD-SDR algorithm, the reduced variables are naturally made available by taking linear combinations of the original predictors. In other words, $\boldsymbol{X}^* = \hat{\boldsymbol{\beta}}^T\boldsymbol{X}$. Any classical regression model could then be applied to the reduced variables.

The dimension-reduction procedure in Algorithm 1 assumes that the structure dimension $d$ (i.e., the number of reduced variables) is known; however, in real scenarios, it must be estimated. In this study, the subspace distance method was adopted from Ye and Weiss (2003) to determine the dimensions of the MD-SDR, and is denoted as $\hat{d}_{mmrn}$. The subspace distance is defined as $\Delta_k(\hat{\mathcal{S}}_k, \hat{\mathcal{S}}_k^b) = ||\mathbf{P}_{\hat{\mathcal{S}}_k} - \mathbf{P}_{\hat{\mathcal{S}}_k^b}||$, where $\hat{\mathcal{S}}_k$ and $\hat{\mathcal{S}}_k^b$ are the estimated subspaces from the original and bootstrapped data sets respectively, $\mathbf{P}_{\hat{\mathcal{S}}}$ is the orthogonal projections onto $\hat{\mathcal{S}}$, and $||\cdot||$ is the maximum singular value of a matrix. The dimension $d$ of the MD-SDR is then estimated by $\hat{d}_{mmrn} = \arg\min_k \Delta_k(\hat{\mathcal{S}}_k, \hat{\mathcal{S}}_k^b)$.Cook (1998) highlighted that sufficient dimension-reduction methods usually reduce the dimension to 1-3. Therefore, it is sufficient to set $k$ from one to ten.

**Table 3** Evaluation measures

| Measures | Formulas |
|---|---|
| Prediction accuracy (PA) | $\frac{TP + TN}{TP + TN + FP + FN}$ |
| False positive rate (FPR) | $\frac{FP}{FP + TN}$ |
| False negative rate (FNR) | $\frac{FN}{FN + TP}$ |
| Precision | $\frac{TP}{TP + FP}$ |
| Recall | $\frac{TP}{TP + FN}$ |
| F1-score | $\frac{2 * Precision * Recall}{Precision + Recall}$ |
| AUC | Area under the ROC curve |

TN, True negative; TP, true positive; FP, false positive; FN, false negative

#### Evaluation measures

To evaluate the performance of the proposed method and compare it with other methods reported in the literature, several commonly used evaluation measures were adopted, as summarized in Table 3. The first column of Table 3 lists the performance measures used, and the corresponding formulas are presented in the second column. The sensitivity between the false-positive and true-positive rates across various classification thresholds was assessed using the receiver operating characteristic (ROC) curve. A quantitative measure related to the ROC curve is the area under the curve (AUC), which is used to measure classification performance. All evaluation measures ranged from 0 to 1, with a higher value indicating better performance in Prediction Accuracy (PA), Precision, Recall, F1-score and AUC whereas higher values indicated worse performance in the False Positive Rate (FPR) and False Negative Rate (FNR).

#### *Case study 1: dimension reduction in* `credit` *data*

The MD-SDR technique was applied to the `credit` dataset introduced in the Data Description subsection. The variable *class* is considered the response and is denoted by $Y$, whereas the remaining 20 variables are considered predictors and are denoted by $X_1, \cdots, X_{20}$. The performance of MD-SDR was compared with that of PCA because PCA is a well-known variable reduction tool. In the dimension reduction application, all discretized variables were treated as numerical, and dummy coding was used for the binary variables. All predictors were standardized to have a mean of zero and a variance of one to ensure a fair comparison, because PCA is sensitive to variable scales.

Parallel to the MD-SDR, the structural dimensions of the reduced variables in the PCA must be determined. The cumulative percentage of variance explained (CPVE) was used by principal components (PCs), and a threshold of 90% was adopted, which has been commonly used in previous studies (Valle et al. 1999; Ulfarsson and Solo 2014).

Using a determined structural dimension, the MD-SDR/PCA technique was applied to the dataset to produce a set of reduced variables. The reduced variables were then compared from MD-SDR and PCA through a regression model by analyzing the prediction power. More specifically, the prediction accuracy is considered from a logistic regression model because the response is binary. The logistic model is expressed by the following equation:

$$\log \frac{P(Y=1)}{P(Y=0)} = \gamma_0 + \gamma_1 X_1^* + \cdots + \gamma_{\hat{d}} X_{\hat{d}}^*, \tag{0.2}$$

where $\gamma_i,\ i = 0, \cdots, \hat{d}$ are the regression coefficients, $X_1^*, \cdots, X_{\hat{d}}^*$ are the $\hat{d}$ reduced variables, $P(Y=1)$ and $P(Y=0)$ are the probabilities of having value 1 and 0 for the response, respectively, and $\log \frac{P(Y=1)}{P(Y=0)}$ is referred to as the log odds. The log odds are the linear functions of the reduced predictors.

The prediction accuracy was calculated using ten-fold cross-validation. The dataset is randomly divided into ten folds of equal size (i.e., each fold has a size of 100), and they are denoted by $D_j,\ j = 1, \cdots, 10$. One fold $D_j$ was selected as the test data, and use the remaining nine folds $D_{-j}$ as the training data to fit the model. Here, $D_{-j}$ denotes the nine folds, excluding the $j$th fold. Based on the reduced predictors in

the test data, the response was predicted using logistic regression coefficients from the training data with a threshold of 0.5, and the predicted response is denoted as $\hat{Y}$. Subsequently, the number of matched cases is recorded and denoted as $m_j$. That is, $m_j = \sum_{i=1}^{100} I(Y_i = \hat{Y}_i)$, where $j = 1, \cdots, 10$ and $I(Y_i = \hat{Y}_i)$ is the indicator function with a value of 1 if $Y_i = \hat{Y}_i$, and 0 otherwise. The process was repeated ten times, with $j$ ranging from 1 to 10, and the overall prediction accuracy was defined as $\text{PA} = \sum_{j=1}^{10} m_j/1000$. A large criterion value was preferred.

Note that the modeling technique was not restricted to logistic regression. Appropriate regressions can be applied to other response types. For example, if the response is continuous, ordinary least squares can be utilized, whereas if the response is ordinal, a cumulative logistic model can be employed.

## Credit scoring

A three-step model was proposed to calculate credit scores. In addition to the dimension reduction and regression model steps, a third step was performed to develop a scorecard using the estimated coefficients. Thus, this new method is named as a dimension reduction assisted credit scoring model (DRA-CS). This procedure is summarized in Fig. 1.

In Step 3, the coefficients estimated from the regression model are used to develop a scorecard, which is standard practice in credit scoring. Because the coefficients from the regression model are designated for the reduced variables (i.e., $\mathbf{X}^*$), they must be transformed back into the original variables (i.e., $\mathbf{X}$). More specifically, $\mathbf{Y} = f(\gamma_0 + \boldsymbol{\gamma}^T\mathbf{X}^*) + \epsilon = f(\gamma_0 + \boldsymbol{\gamma}^T\boldsymbol{\beta}^T\mathbf{X}) + \epsilon = f(\gamma_0 + \boldsymbol{\xi}^T\mathbf{X}) + \epsilon$. Here, $\boldsymbol{\gamma}$ is the vector of coefficients corresponding to the reduced variables and $\boldsymbol{\xi} = \boldsymbol{\beta}\boldsymbol{\gamma}$ is the vector of coefficients corresponding to the original variables, which is a product of the reduced coefficients and the directions to reduce the variables. Then, the coefficient $\hat{\xi}(j,k)$ is obtained for each variable $X_j$, $j = 1, \ldots, J$, and category $k$, $k = 1, \ldots, (K-1)$. For categorical or discretized variables with $K$ categories, because one category was omitted and used as a base (reference) level, the coefficient associated with this category was set to zero. That is, $\hat{\xi}(j,K) = 0$. Let $N(j, k)$ denote the number of credit points associated with category $k$ in variable $X_j$ Then, it followsthat

$$N(j,k) = 100 \times \frac{\hat{\xi}(j,k) - min_K[\hat{\xi}(j,1), \ldots, \hat{\xi}(j,K)]}{\sum_l max_M[\hat{\xi}(l,1), \ldots, \hat{\xi}(l,M)] - min_M[\hat{\xi}(l,1), \ldots, \hat{\xi}(l,M)]} \tag{0.3}$$

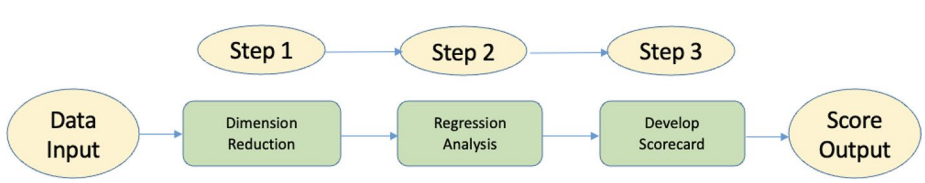


**Fig. 1** Flowchart for the proposed three-step credit scoring model

where the denominator represents the sum of the largest differences between the coefficients of the same variable, calculated across all variables. When all $\hat{\xi}(j,1),\ldots,\hat{\xi}(j,(K-1)) \geq 0$, then $min_K[\hat{\xi}(j,1),\ldots,\hat{\xi}(j,K)] = 0$, and when some of the $\hat{\xi}(j,1),\ldots,\hat{\xi}(j,(K-1)) < 0$, then $min_K[\hat{\xi}(j,1),\ldots,\hat{\xi}(j,K)] < 0$. Note: Eq. (0.3) was adopted from Tufféry (2011) and modified to be more comprehensible. The denominator captures the overall difference between the largest and smallest coefficients across all variables, whereas the numerator measures the difference between each coefficient and the smallest coefficient within a certain variable. The advantage of this formulation is that the sum of the maximum scores of the variables is 100, which facilitates easy interpretation.

#### *Case Study 2: Credit Scoring by the Three-step approach in* `credit` *data*

The MD-SDR method with logistic regression was used to develop a credit scorecard for the `credit` dataset. The credit scorecards from the MD-SDR were compared to those obtained using the PCA method and stepwise model selection approach. Tufféry (2011) used the stepwise model selection approach with the logistic regression in the computation of the credit risk scores for the `credit` data, which provides an excellent baseline for comparison to the MD-SDR results. The stepwise regression approach consists of the step-by-step iterative construction of a regression model that involves the selection of independent predictors for use in the final model. This process involves adding or removing potential predictors in succession, and testing for statistical significance after each iteration. To ensure a fair comparison, the proposed three-step approach was also applied to the PCA. See Figure. 2 for a detailed flowchart.

To obtain the coefficient for each level of the categorical variables, as in the stepwise model selection approach, dummy variables were created to model the categorical variables in the MD-SDR and PCA methods. In the case of categorical variables with multiple levels, the base level was selected as follows Tufféry (2011): This allowed us to compare our credit score results from the MD-SDR approach with those obtained using existing approaches. Note that data handling within the MD-SDR approach for credit score modeling is different from the dimension reduction application, where

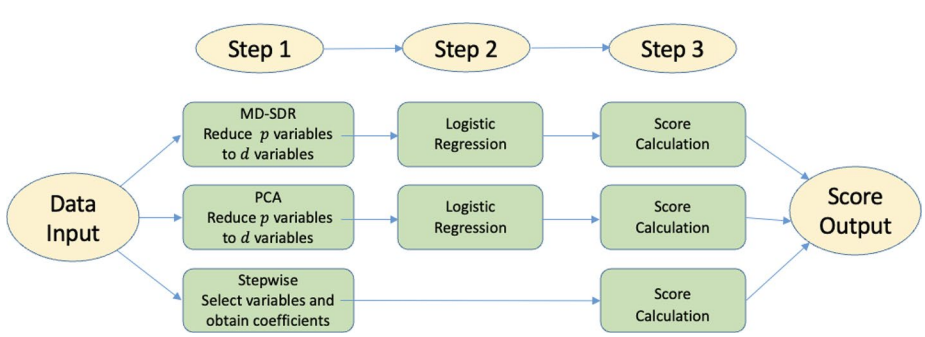


**Fig. 2** Flowchart for the three credit scoring models

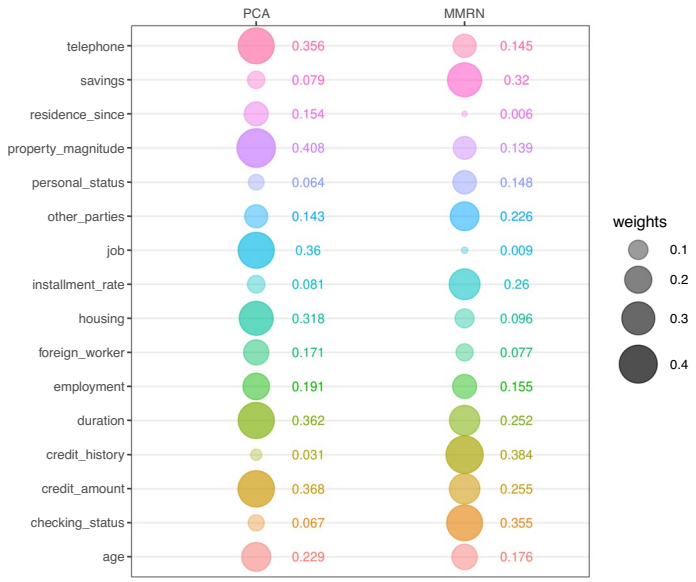


**Fig. 3** Comparison of the weights generated by the PCA and MD-SDR methods for the first 16 variables. The size of the bubble increases with the magnitude of weight

**Table 4** Eigenvalues distribution with the corresponding CPVEs obtained by the twenty principal components from PCA

| **PC** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** |
|---|---|---|---|---|---|---|---|---|---|---|
| Eigenvalue | 49.35 | 42.75 | 37.50 | 36.24 | 34.46 | 33.74 | 32.74 | 32.60 | 31.13 | 30.10 |
| CPVE | 0.08 | 0.15 | 0.21 | 0.27 | 0.32 | 0.38 | 0.43 | 0.49 | 0.54 | 0.58 |
| **PC** | **11** | **12** | **13** | **14** | **15** | **16** | **17** | **18** | **19** | **20** |
| Eigenvalue | 29.39 | 29.13 | 28.65 | 27.48 | 26.72 | 26.13 | 23.73 | 22.72 | 22.25 | 20.06 |
| CPVE | 0.63 | 0.68 | 0.73 | 0.77 | 0.81 | 0.86 | 0.89 | 0.93 | 0.97 | 1.00 |

the variables are simply discretized (case study 1). When the dummy variables are combined, there are 59 variables as shown in Table 7. After employing the MD-SDR method, the original dataset size of $1000 \times 59$ was reduced to the size $1000 \times 1$, producing a single dimension. The PCA method reduced the original dataset to a smaller size of $1000 \times 29$, producing 29 principal components with more than 90% of the explained CPVE.

The comparison can also be viewed from another perspective: the Stepwise approach is a model-based variable selection technique, PCA is an unsupervised dimension-reduction method, and MD-SDR is a supervised model-free dimension-reduction approach.

## Application and results

This section presents the results of the analysis, which includes two parts: application to dimension reduction and credit score modeling. All research questions will be addressed.

**Table 5** Subspace distance between the estimated subspaces from the original and bootstrapped data set for each different *k* obtained by the MD-SDR method

| *k* | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Subspace distance | 0.00 | 0.63 | 0.60 | 0.76 | 0.66 | 0.78 | 0.84 | 0.90 | 0.87 | 0.88 |

### Case study 1 results

By applying the MD-SDR approach to the `credit` data, its performance was compared with that PCA in several components. Initially, to achieve the first research goal, the absolute weights were compared in the first direction (i.e., $|\boldsymbol{\beta}_{1j}|, j = 1, \cdots, 20$) using a bubble plot. A set of ten variables was selected with the largest absolute weights from both the MD-SDR and PCA, and the union of the two sets was then determined, yielding 16 variables with the weights given in Fig. 3. The size of the bubbles increases with the magnitude of the weight. Significant differences were observed in the weights between the two methods across all variables. The PCA method identifies the most important predictors with the largest weights as property magnitude, credit amount, and duration, whereas the most important variables from the MD-SDR method are credit history, status checking, and savings. Note that a comparison using the weights makes sense because both directions from the MD-SDR and PCA have unit lengths.

The second comparison between MD-SDR and PCA is based on the degree of reduction (i.e., number of reduced variables $\hat{d}$). According to the results reported in Table 4, the CPVE reached 90% for the first 18 PCs. Therefore, in PCA, 18 PCs are required to maintain a reasonable portion of the information after reducing the variable dimensions (i.e., $\hat{d}_{pca} = 18$).

The results presented in Table 5 show that in the MD-SDR approach, the subspace distance (a dimension selection criterion for MD-SDR) is minimized when $k = 1$. This means that reducing the data to only one variable (a linear combination of all the original predictors) is sufficient to capture all the regression information between the response variable and predictors. In other words, with the MD-SDR, a single reduced variable determined by the subspace distance criterion can effectively summarize the regression relationship between the response and predictors. Although reducing the data to one dimension may seem extreme, it is a common practice in sufficient dimension reduction (SDR) and has been widely studied in previous research.

**Table 6** Summary of the prediction accuracy across three methods, the two PCA methods and the MD-SDR method by ten-fold cross-validation with the estimated dimension $\hat{d}$

| Methods | MD-SDR ($\hat{d}_{mmrn} = 1$) | PCA($\hat{d}_{pca} = 1$) | PCA ($\hat{d}_{pca} = 18$) |
|---|---|---|---|
| PA | 0.76 | 0.70 | 0.75 |
| FPR | 0.10 | 0.00 | 0.10 |
| FNR | 0.58 | 1.00 | 0.59 |
| Precision | 0.64 | NA | 0.63 |
| Recall | 0.42 | 0 | 0.41 |
| F1-score | 0.51 | NA | 0.50 |
| AUC | 0.66 | 0.50 | 0.65 |

**Table 7** Credit scores from the different methods: SW, PCA, and MD-SDR

| | | Credit points | | |
|---|---|---|---|---|
| **Variable** | **Class** | **SW** | **PCA** | **MD-SDR** |
| Age | ≤ 25 years | 8 | 4 | 5 |
| | > 25 years* | 0 | 0 | 0 |
| Credit_duration | ≤ 15 months* | 0 | 0 | 0 |
| | 16 − 36 months | 9 | 3 | 4 |
| | > 36 months | 13 | 7 | 5 |
| Amount | < 4000 | / | 0 | 0 |
| | ≥ 4000* | / | 6 | 4 |
| Accounts | No checking account* | 0 | 0 | 1 |
| | ≥ 200 euros | 13 | 3 | 0 |
| | 0 − 200 euros | 19 | 9 | 11 |
| | < 0 euros | 25 | 11 | 14 |
| Savings | No savings or > 500 euros* | 0 | 0 | 0 |
| | < 500 euros | 9 | 5 | 4 |
| Guarantees | Guarantor | 0 | 0 | 0 |
| | No guarantor* | 18 | 2 | 7 |
| Other_credits | No credit | 0 | 0 | 0 |
| | Other bank or institutions* | 6 | 3 | 3 |
| Credit_history | No credit at anytime* | 0 | 0 | 0 |
| | Credits without delay | 9 | 6 | 4 |
| | Credits with non-payments | 25 | 10 | 10 |
| Purpose | New vehicle | / | 13 | 10 |
| | Used vehicle | / | 0 | 0 |
| | Home | / | 7 | 5 |
| | Education/business | / | 8 | 4 |
| | Others* | / | 6 | 1 |
| Housing | Owner* | / | 0 | 0 |
| | Rented flat/free apartment | / | 4 | 2 |
| Employment | > 4 years* | / | 0 | 0 |
| | Unemployment/< 1year | / | 3 | 4 |
| | 1 − 4 years | / | 2 | 2 |
| Property_magnitude | No known asset* | / | 2 | 5 |
| | Other asset | / | 2 | 2 |
| | Property | / | 0 | 0 |
| Dependents | 0 − 2* | / | 0 | 0 |
| | ≥ 3 | / | 2 | 0 |
| Installment_rate | < 20* | / | 5 | 2 |
| | > 35 | / | 0 | 1 |
| | 25 − 35 | / | 1 | 0 |
| | 20 − 25 | / | 3 | 1 |
| Personal_status | male:married/widowed* | / | 1 | 0 |
| | Male:divorced/separated | / | 5 | 3 |
| | Female:divorced/separated/married | / | 2 | 0 |
| | Male:single | / | 0 | 0 |
| Residence | > 7 years* | / | 4 | 3 |
| | < 1year | / | 0 | 0 |
| | 1 − 4 years | / | 6 | 6 |
| | 4 − 7 years | / | 2 | 2 |

**Table 7** (continued)

| Variable | Class | Credit points | | |
|---|---|---|---|---|
| | | SW | PCA | MD-SDR |
| Job | executive/self-employment/higher civil servant* | / | 1 | 4 |
| | Unemployment/unskilled with no permanent residence | / | 2 | 0 |
| | Unskilled with permanent residence | / | 0 | 2 |
| | Skilled worker/minor civil servant | / | 0 | 4 |
| Existing_credit | > 6* | / | 1 | 5 |
| | 1 | / | 2 | 0 |
| | 2 − 3 | / | 5 | 1 |
| | 4 − 5 | / | 0 | 5 |
| Telephone | None* | / | 1 | 1 |
| | Yes | / | 0 | 0 |
| Foreign_worker | No * | / | 0 | 0 |
| | Yes | / | 4 | 6 |

*Indicates the base level

In SDR, several reduced variables are selected based on specific criteria tailored for the dimension reduction method, such as minimizing the subspace distance for the DCOV-based SDR approach (MD-SDR). This criterion ensures that the essential information required for the regression analysis is retained. However, PCA reduces the original predictors to a smaller number of components by preserving most variations in the predictors. However, PCA does this without considering the response variable, which is the key difference from the MD-SDR approach.

In conclusion, the MD-SDR achieves a greater reduction than PCA. The additional gain obtained by reducing the dimensions from 20 to 1 suggests the ability to investigate the relationship between the response and reduced variables through a visual method. For instance, one can look at a scatterplot of the response and the reduced variable (summary plot).

The third comparison was conducted using the predictive power of two-dimensional reduction techniques. The results from the MD-SDR with $\hat{d}_{mmrn} = 1$ and the PCA with $\hat{d}_{pca} = 1$ and $\hat{d}_{pca} = 18$ are listed in Table 6. Note that the result of the PCA with $\hat{d}_{pca} = 1$ is included here for comparison with MD-SDR under the same degree of reduction. From Table 6, it is observed that with the same degree of reduction (reducing the predictor dimension from 20 to $\hat{d} = 1$), the MD-SDR method outperforms PCA by 8.5% (relative increase in terms of the prediction accuracy). If the estimated dimension is increased from $\hat{d}_{pca} = 1$ to $\hat{d}_{pca} = 18$, the performance of PCA increases, and the performance is similar to that of MD-SDR. However, an increase in the estimated dimension eliminates the ability to visually display the relationship between the response and reduced variables because it is challenging to draw plots with more than three dimensions. The prediction performance was also evaluated by FPR, FNR, Precision, Recall and F1-score. Across all evaluation measures, the proposed MD-SDR was better than PCA with both 1 and 18 dimensions. It should be noted that when using one dimension, the PCA classified all cases as zero; thus, the precision and F1-score were undefined. While the FNR appears to be somewhat high for PCA and MD-SDR methods, this is

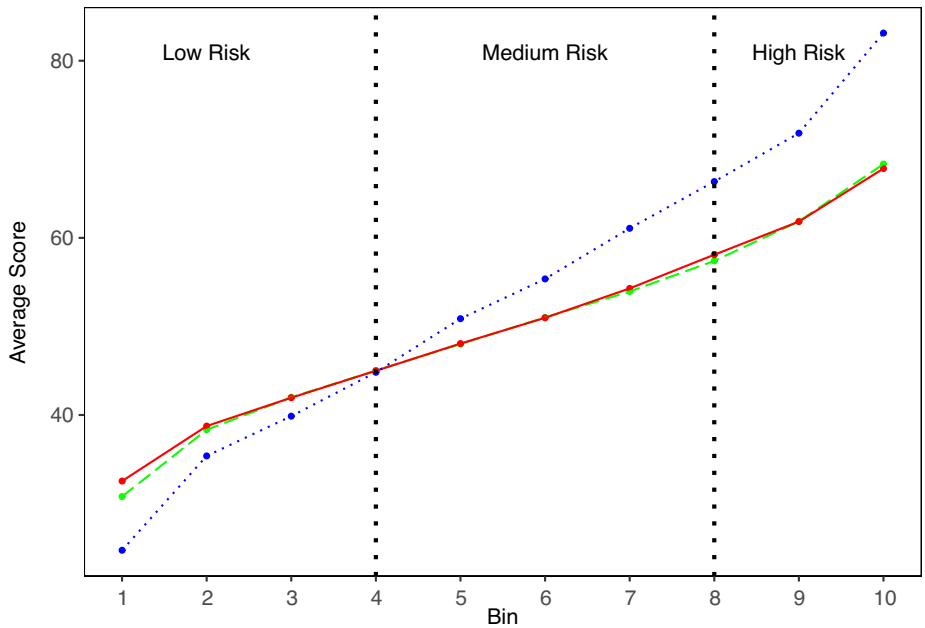


**Fig. 4** Average scores by bin across the three methods. Dotted blue color line (SW), longdash green color line (PCA), and solid red color line (MD-SDR). The region of low risk is labeled for bins 1-4, medium risk for bins 5-8, and high risks for bins 9-10

expected considering the significantly larger proportion (70%) of good risk in the data. The three comparisons in this section help to answer the third research question.

### Case study 2 results

To answer the second research question, this study follows the procedures introduced in the Data and Methodology section and presents the scorecard created in Table 7, where * denotes the base levels. Table 7 illustrates the scorecard based on all variables used across the three methods: stepwise regression (SW), PCA, and MD-SDR. Since a large number of variables are eliminated using the SW method, for these variables, the number of points are not available. One drawback of the SW method is that it omits several important characteristics of an applicant's profile. The other two methods, PCA and MD-SDR, allow us to convert the resulting directions to the original variables to compute a more comprehensive credit score.

To compare the three scorecards, points were assigned to each applicant included in the `credit` data in Table 7. The points were then divided for each method into 10 deciles (or 10 bins) and further into three classes (i.e., low risk, medium risk, and high risk). The higher the score, the higher is the risk. For example, if an applicant's score falls into the high-risk class, the financial institution will decide to reject the application for loans or insurance considerations. If the applicant's score was within the first four bins, the applicant would be easily approved for loan or insurance considerations. Applications classified as medium-risk are somewhere in the middle and may be accepted or rejected depending on the underwriting guidelines of the institution. Differences in

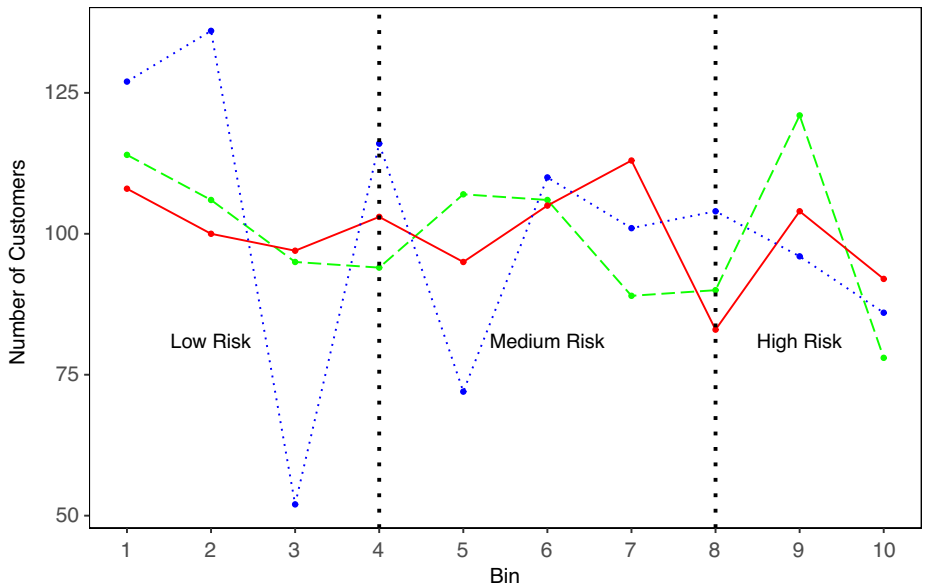


**Fig. 5** Number of customers by bin across the three methods. Dotted blue color line (SW), longdash green color line (PCA), and solid red color line (MD-SDR). The region of low risk is labeled for bins 1-4, medium risk for bins 5-8, and high risks for bins 9-10

credit score calculations have a significant impact on applicants applying for loans or insurance.

Figure 4 shows the average score for each bin for the three methods used to calculate the credit scores. Areas of low, medium, and high risk are labeled in the plot. The results of the SW method deviate significantly from the other two methods, showing lower

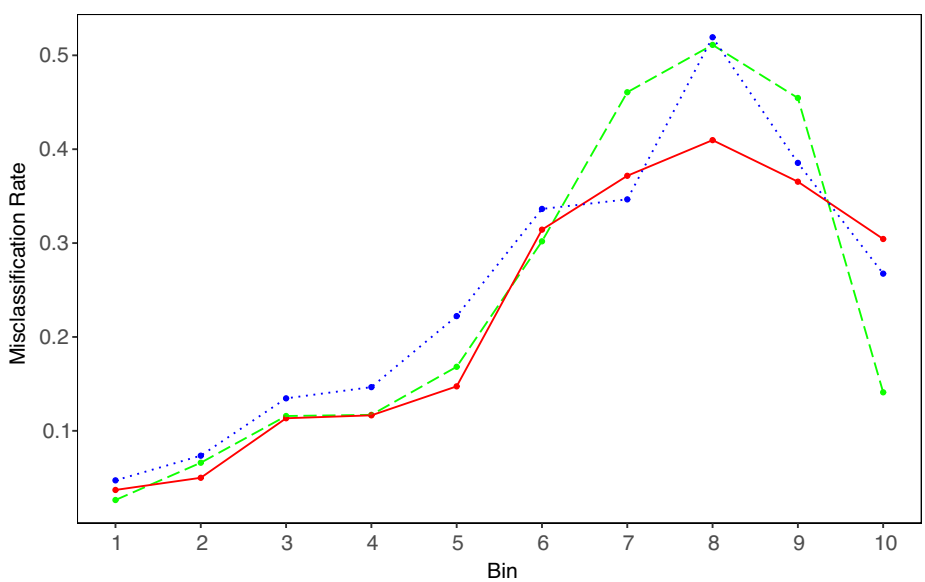


**Fig. 6** Misclassification rate. The proportion of applicants misclassified in each bin across the three methods. Dotted blue color line (SW), longdash green color line (PCA), and solid red color line (MD-SDR)

average scores in the first four bins and higher average scores in the last six bins, when compared to the other two methods. For example, a customer with an average score of 60 would be classified as high risk using the PCA and MD-SDR methods. However, the same customer was classified as having medium risk using the SW method. The PCA and MD-SDR methods are more sensitive for detecting high risks. The average PCA and MD-SDR scores were similar across all the bins.

The variation in the number of customers by bin for all three methods is shown in Fig. 5. The highest variability was observed using the SW method. In the SW method, some bins have a very small number of customers, whereas others have a very large number. For the first five bins, these numbers were outside the range of the numbers calculated using the PCA and MD-SDR methods. Looking at the high-risk regions, it was observed that the number of high-risk applicants was higher for the PCA method than for the MD-SDR and SW methods. Because high-risk applicants are also those who are most likely to be rejected for loans or insurance considerations, these decisions have a financial impact on both applicants and lenders. To estimate the misclassification rates for good and bad risks across the three methods, a comparison was performed between the predicted risk classes of each applicant generated by the SW, PCA, and MD-SDR methods and that of the risk class reported in the consumer `credit` data. The misclassification rate represents both Type I (rejecting a good risk) and Type II (failing to reject a bad risk) errors. Ideally, the misclassification rate should be minimized. However, it appears that the rate increases from the lower to higher bins. In the textttcreditdata file, 70 % were classified as good risk (creditworthy or history of no non-payments), and 30 % were identified as bad risk (not creditworthy or history of existing non-payments). A split of 70:30 was used to divide the predicted credit scores into good and bad risk categories across all three methods, and then the proportion of misclassified applicants in each of the ten bins was computed. The results in terms of misclassfication rate are shown in Figure 6. The misclassification rate is defined as the ratio of falsely predicted cases to the total number of cases. In other words, $\frac{FP+FN}{TP+TN+FP+FN}$. On average, the MD-SDR method generated a lower misclassification rate, except for bin 10. In conclusion, the MD-SDR method was superior to the PCA and SW methods. In addition, this study argues that both the PCA and MD-SDR methods are preferred over the SW method because the SW method uses partial customer information. When the PCA and MD-SDR methods are used for dimension reduction, the number of points can be computed for all the original variables, which allows us to include full information about each consumer in the database while still performing a dimension reduction of the data. For Bins 7-9, the PCA method had a significantly higher misclassification rate than the MD-SDR method. The scores allocated to Bins 7-9 may also represent consumers on the borderline between good and bad risk. The highest misclassification rate was reported for Bin 8. Those borderline cases may be rejected by one lender but accepted by another lender. The risks associated with these misclassification rates have financial consequences for the profitability and/or losses of financial institutions.

## Sensitivity analysis and robustness testing

In this section, we discuss the sensitivity and robustness of the proposed method. First, a sensitivity analysis was performed through a simulation study, which is an essential tool for gauging the performance of the proposed method and comparing the performances of

alternative methods under different settings. Next, we evaluate the out-of-sample robustness of the above results across two popular datasets previously considered by Dumitrescu et al. (2022) in credit-scoring research.

### Sensitivity analysis

Sensitivity analysis of the MD-SDR was conducted under various scenarios, and the numerical results were compared with those from the PCA (with many directions) and PCA1 (with only one direction) methods. In this simulation study, cases were artificially created in which the responses and predictors were related to different degrees, such as strong/moderate/weak signals. It is important to note that the methodology used for generating artificial datasets is a contribution of this study.

The artificial datasets were generated using the following model:

$$P(\mathbf{Y} = 1) = \frac{1}{1 + e^{-\mathbf{X}^T \boldsymbol{\beta}}},$$

where $\mathbf{X} \in R^{p_x} \overset{iid}{\sim} \text{Poisson}(\lambda)$ is the predictor matrix, $\boldsymbol{\beta}$ is the $p_x \times 1$ coefficient matrix, and $\mathbf{Y}$ is the response that is generated based on a Bernoulli distribution and the contaminate threshold $S = (s_l, s_u)$. The choice of the Poisson distribution for the predictors

**Table 8** Summary of the prediction accuracy results for six different settings of the contaminate threshold $S = \{(0, 1), (0.1, 0.9), (0.2, 0.8), (0.3, 0.7), (0.4, 0.6), (0.5, 0.5)\}$ with a sample size $N = 1000$

| (*S*) | (0, 1) | (0.1, 0.9) | (0.2, 0.8) | (0.3, 0.7) | (0.4, 0.6) | (0.5, 0.5) |
|---|---|---|---|---|---|---|
| (*k*) | PA results | | | | | |
| MD-SDR | **0.68** | **0.68** | **0.71** | **0.77** | **0.91** | **0.99** |
| PCA1 | 0.55 | 0.55 | 0.56 | 0.58 | 0.59 | 0.62 |
| PCA | 0.67 | 0.67 | 0.70 | 0.76 | 0.88 | 0.93 |
| | FPR results | | | | | |
| MD-SDR | **0.32** | **0.32** | **0.29** | **0.23** | **0.09** | **0.02** |
| PCA1 | 0.46 | 0.46 | 0.45 | 0.43 | 0.40 | 0.42 |
| PCA | 0.33 | 0.33 | 0.30 | 0.24 | 0.12 | 0.07 |
| | FNR results | | | | | |
| MD-SDR | **0.32** | **0.32** | **0.29** | **0.23** | **0.09** | **0.01** |
| PCA1 | 0.45 | 0.45 | 0.44 | 0.43 | 0.42 | 0.41 |
| PCA | 0.33 | 0.33 | 0.30 | 0.24 | 0.12 | 0.07 |
| | Precision results | | | | | |
| MD-SDR | **0.68** | **0.68** | **0.71** | **0.77** | **0.91** | **0.99** |
| PCA1 | 0.55 | 0.55 | 0.57 | 0.58 | 0.59 | 0.61 |
| PCA | 0.67 | 0.67 | 0.70 | 0.76 | 0.88 | 0.93 |
| | Recall results | | | | | |
| MD-SDR | **0.68** | **0.68** | **0.71** | **0.77** | **0.91** | **0.99** |
| PCA1 | 0.55 | 0.55 | 0.56 | 0.57 | 0.58 | 0.59 |
| PCA | 0.67 | 0.67 | 0.70 | 0.76 | 0.88 | 0.93 |
| | F1-score results | | | | | |
| MD-SDR | **0.68** | **0.68** | **0.71** | **0.77** | **0.91** | **0.99** |
| PCA1 | 0.54 | 0.54 | 0.56 | 0.57 | 0.58 | 0.58 |
| PCA | 0.67 | 0.67 | 0.70 | 0.76 | 0.88 | 0.93 |

The best results are presented in boldface

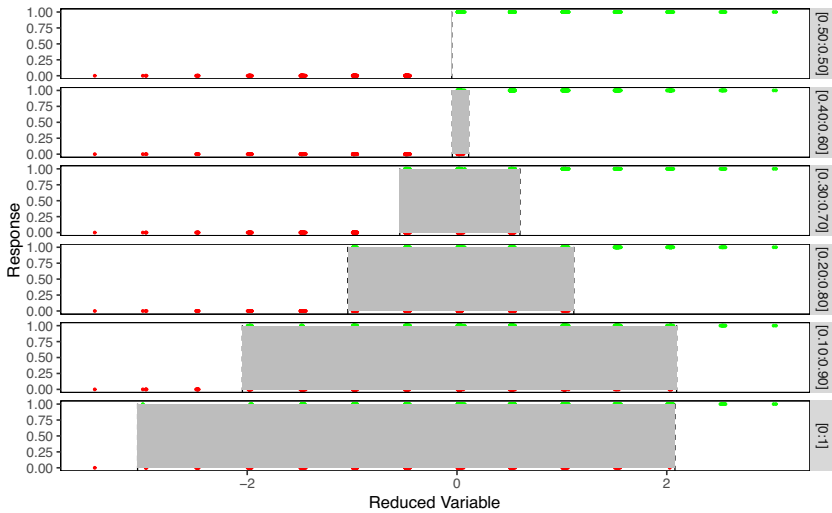


**Fig. 7** Summary plot of the six different contaminate thresholds. The contaminate level is increasing from the top plot to the bottom plot. The shaded region on each plot indicates the range of reduced variable values for which both response variable values (0 and 1) are possible. Red colored points correspond to $\mathbf{Y} = 1$, while the green colored points correspond to $\mathbf{Y} = 0$

was aimed at assessing the resilience of the proposed model when confronted with non-normally distributed data. A Bernoulli distribution was adopted to classify customers into good or bad credit categories, representing two possible outcomes. Furthermore, $P(\mathbf{Y} = 1)$, the probability of success in the Bernoulli process, is defined specifically as

$$\mathbf{Y} = \begin{cases} 0 & P(\mathbf{Y} = 1) \leq s_l \\ Bernoulli(P(\mathbf{Y} = 1)) & s_l < P(\mathbf{Y} = 1) < s_u \\ 1 & P(\mathbf{Y} = 1) \geq s_u. \end{cases}$$

The Bernoulli random variable artificially induces contamination because $\mathbf{Y}$ can be either zero or one based on the probability $P(\mathbf{Y} = 1)$. In addition, the larger the interval width in $S$, the greater the contamination of the data. The simulation settings were as follows: sample size $N = 1000$, predictor dimension $p_x = 10$, parameter for the Poisson distribution $\lambda = 1$, coefficient $\boldsymbol{\beta} = (0.5, -0.5, 0.5, -0.5, 0, 0, 0, 0, 0, 0)^T$, and six different settings for the contaminant threshold $S \in \{(0, 1), (0.1, 0.9), (0.2, 0.8), (0.3, 0.7), (0.4, 0.6), (0.5, 0.5)\}$. These six settings correspond to scenarios with different levels of relationships between $\boldsymbol{Y}$ and $\mathbf{X}$. Large contamination corresponds to a weak relationship between $\boldsymbol{Y}$ and $\mathbf{X}$. In this sensitivity analysis, the same procedure was repeated 1000 times and the evaluation measures were recorded for each repetition. A summary of the averaged evaluation measures from 1000 repetitions corresponding to these six scenarios is presented in Table 8.

From Table 8, it can be observed that the PA results from MD-SDR are uniformly better than those PCA in both the one- and multi-dimensional cases. When the relationship between the response and predictor was strong, the difference between the MD-SDR and PCA in terms of the PA results became more evident. In addition, as the contamination rate decreased (i.e., the relationship became stronger), the performance of all three

**Table 9** Subspace distance between the estimated subspaces from the original and bootstrapped data set for each different *k* obtained by the MD-SDR method for HMEQ Data

| *k* | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Subspace distance | 0.00 | 0.45 | 0.80 | 0.85 | 0.42 | 0.21 | 0.32 | 0.34 | 0.45 | 0.19 |

**Table 10** Subspace distance between the estimated subspaces from the original and bootstrapped data set for each different *k* obtained by the MD-SDR method for ACA data

| *k* | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Subspace distance | 0.00 | 0.85 | 0.92 | 0.95 | 0.98 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |

methods improved. The same conclusion applies to all other evaluation measures. Note that some of the results may appear the same because of rounding.

Figure 7 shows a summary plot of the six contaminant threshold value settings. These plots are ordered based on the value of the contaminant threshold, increasing from top to bottom. The range of negative values of the reduced variable corresponds to a response value equal to zero, $\mathbf{Y} = 0$ (red), whereas the positive range of the values associated with the reduced variable corresponds to a response value equal to 1, $\mathbf{Y} = 1$ (green).

The shaded region in each plot indicates the range of reduced variables for which both $\mathbf{Y} = 0$ and $\mathbf{Y} = 1$ are possible. As the contamination threshold increases, the shaded area also increases, which means that there is less space for a 1-to-1 relationship between the reduced variable and the response variable. For example, when the contamination threshold is (0.5, 0.5), there is a 1-to-1 relationship between the reduced variable and response variable, and no shaded region exists. However, when the contamination threshold setting was (0.30, 0.70), the shaded region was very large, with very little space for the 1-to-1 relationship between the reduced and response variables. This phenomenon was most pronounced when the contaminant threshold was set to the maximum value of (0, 1). The MD-SDR method exhibited the highest PA values relative to the other two PCA methods for all settings.

Although MD-SDR shows better estimation results, PCA is faster than MD-SDR in the dimension-reduction procedure. To estimate the directions for reducing the data size, PCA applies eigen-decomposition to the covariance matrix directly, whereas MD-SDR involves double iterations and tuning parameter determination. Iterations and parameter determination may lead to significantly higher computational costs than in PCA.

### Robustness testing

Out-of-sample robustness testing of the proposed method is conducted across two additional popular datasets: Home Equity(HMEQ) data and Australian Credit Approval(ACA) data. The ACA dataset, which focuses on credit card applications, originates from the University of California at Irvine (UCI) and was provided by Quinlan. It was previously employed as a credit-approval database for the Statlog project. Initiated by the European Commission in the 1990 s, the Statlog project aims to promote R&D in statistical learning and data mining. It involves creating benchmark

**Table 11** Summary of the prediction accuracy across three methods, the two PCA methods and the MD-SDR method by ten-fold cross-validation with the estimated dimension $\hat{d}$ for HMEQ data

| Methods | MD-SDR ($\hat{d}_{mmrn} = 1$) | PCA($\hat{d}_{pca} = 1$) | PCA ($\hat{d}_{pca} = 14$) | MR ($d_{MR} = 1$) |
|---|---|---|---|---|
| PA | 0.84 | 0.80 | 0.84 | 0.83 |
| FPR | 0.03 | 0.00 | 0.03 | 0.04 |
| FNR | 0.69 | 1.00 | 0.70 | 0.70 |
| Precision | 0.71 | NA | 0.72 | 0.68 |
| Recall | 0.31 | 0.00 | 0.30 | 0.30 |
| F1-score | 0.43 | NA | 0.43 | 0.42 |
| AUC | 0.64 | 0.50 | 0.64 | 0.63 |

datasets and organizing challenges to encourage researchers to develop and compare machine learning algorithms (see King et al. (1995)). These datasets cover various domains, including medical diagnosis, credit scoring, and pattern recognition. For a detailed summary of the variable descriptions for the ACA data and the HMEQ dataset, refer to Dumitrescu et al. (2022). The authors used these datasets to test the performance of their proposed credit scoring method called penalized logistic tree regression (PLTR), which uses information from decision trees to enhance the performance of logistic regression.

The HMEQ data, also recognized as the housing dataset, are accessible in the SAS library and have been used by several authors as illustrative examples (see Matignon (2007) and Dumitrescu et al. (2022)). Comprising 5960 loans, with 1189 defaults, the data set yields a prior default rate of 19.95%. It encompasses 12 explanatory variables, two of which are nominal. The nominal variables are further coded as dummy variables, which expand the total number of predictors to 16: The same strategies were adopted from Dumitrescu et al. (2022) to impute missing values by replacing the NA's in the categorical variables with the mode and numerical variables with the mean.

The ACA dataset includes six numerical and eight nominal predictors, comprising 690 instances with 307 defaults and a prior default rate of 44.49%. To safeguard data confidentiality, all attribute names and values have been replaced with meaningless symbols. This dataset stands out owing to its diverse mix of attributes, including continuous, nominal with a small number of values, and nominal with a large number of values. In

**Table 12** Summary of the prediction accuracy across three methods, the two PCA methods and the MD-SDR method by ten-fold cross-validation with the estimated dimension $\hat{d}$ for ACA data

| Methods | MD-SDR ($\hat{d}_{mmrn} = 1$) | PCA($\hat{d}_{pca} = 1$) | PCA ($\hat{d}_{pca} = 22$) | MR ($d_{MR} = 1$) |
|---|---|---|---|---|
| PA | 0.86 | 0.80 | 0.86 | 0.87 |
| FPR | 0.20 | 0.14 | 0.14 | 0.16 |
| FNR | 0.07 | 0.27 | 0.13 | 0.09 |
| Precision | 0.79 | 0.80 | 0.83 | 0.82 |
| Recall | 0.93 | 0.72 | 0.87 | 0.91 |
| F1-score | 0.85 | 0.76 | 0.85 | 0.86 |
| AUC | 0.86 | 0.79 | 0.86 | 0.87 |

addition, a few missing values were replaced with the means of the predictive variables. Similar to the HMEQ dataset, dummy coding was applied to handle the categorical variables in ACA dataset which expanded the predictor size to 26. Categories that have few cases and might lead to convergence issues are merged with other categories.

Methodologies similar to those employed for the German dataset were used for these two datasets. To assess the reduced variables for MD-SDR, subspace distances were computed for both the HMEQ and ACA datasets. We present the outcomes in Tables 9 and 10. Both tables illustrate that the subspace distance criterion reaches its minimum value at $k = 1$. This signifies that employing the MD-SDR approach requires only one reduced variable for both datasets. According to the CPVE for PCA, the HMEQ data required 14 principal components (PCs) to attain 90% variation, whereas the ACA dataset required 22 PCs.

Once the required number of reduced variables was determined, a standard logistic regression model was applied to the reduced datasets. Similar to the German credit dataset, ten-fold cross-validation was used to compute the prediction metrics. The results are displayed in Tables 11 and 12 for the HMEQ and ACA datasets respectively. In both tables, the suggested MD-SDR approach demonstrated comparable or superior performance to PCA, whether using one or more PCs. These results align consistently with the observations made regarding the German credit dataset.

The dimension reduction methods generate linear combinations of the original predictors as reduced variables, and similar linear combinations can also be obtained from multiple regression (MR). Regular multiple linear regression is a dimension-reduction approach and an SDR method when the predictor vector is normally distributed (Duan and Li 1991). Therefore, in this robustness test, the results from multiple regression as the dimension-reduction method were also included for comparison. It should be noted that multiple regression can only reduce the original predictors to one reduced variable, making it unnecessary to evaluate the dimensions. Thus, $d_{MR}$ is used instead of $\hat{d}_{MR}$. From a comparison with the MR approach, the proposed MD-SDR is comparable or superior, particularly for the HMEQ data (see Tables 11 and 12 ).

The reader is referred to the Appendix for a comparative analysis with state-of-the-art methods summarized in Dumitrescu et al. (2022).

## Discussion and conclusion

This study introduces a three-step credit scoring model, the DRA-CS, comprising dimension reduction, logistic regression, and scorecard components. The first step employs a novel MMRN algorithm to efficiently solve a distance-based sufficient dimension reduction problem, particularly for high-dimensional data. The second step utilized a logistic regression model for risk classification, producing coefficients for each factor. In the final step, customer credit scores were calculated and used for classification. DRA-CS outperforms current methods such as PCA in terms of dimension reduction and is recommended for high-dimensional datasets. The integration of the MD-SDR component into alternative credit scoring models that consider social and psychological customer information could enhance their performance (see Woo and Sohn (2022)).

To the best of our knowledge, the DRA-CS approach stands out in the credit scoring field for its incorporation of innovative MD-SDR components. This method offers

simplicity in application, efficiency with the MMRN algorithm, model-free supervised reduction, and superior dimension reduction compared with other techniques. MD-SDR effectively extracts essential information from high-dimensional data, reducing dimensions to 1-3. In a sensitivity analysis conducted through a simulation study under various settings, MD-SDR demonstrated superior prediction accuracy compared to PCA. Remarkably, DRA-CS exhibited lower misclassification rates than PCA. Out-of-sample robustness testing was conducted across two additional datasets, revealing that MD-SDR exhibited comparable or superior performance to PCA, whether one or more determined principal components were used.

The DRA-CS method offers the advantage of employing a traditional statistical model such as logistic regression, which is the standard choice for analyzing discrete binary outcome variables. However, an alternative model could replace logistic regression while retaining MD-SDR as the initial step. For instance, deep neural networks (DNN) and other deep learning approaches can benefit from an enhanced feature selection method, such as the MD-SDR approach, to improve learning with denoised data. The growing popularity of DNN and other artificial intelligence algorithms for credit scoring is evident (see Gunnarsson et al. (2021)).

The proposed method in the dimension reduction application considers a linear combination of all predictors; it is unfair to compare variable selection methods that remove several variables completely. In addition, while we agree that eliminating irrelevant variables could enhance interpretability, another motivation for choosing PCA instead of variable selection techniques is that our study focuses primarily on categorizing customers into different credit categories (good or bad) rather than interpreting individual variables.

Apart from comparing dimension reduction methods, another comparison focused on modeling to create a credit scorecard. In this study, a logistic regression was employed, aligned with the credit scoring literature, using the German credit dataset. This choice allowed for a natural comparison among the proposed MD-SDR, PCA, and stepwise logistic regression (a benchmark from the literature). Although logistic regression is used here, it can be substituted with other state-of-the-art modeling methods in the credit scoring domain, such as support vector machines, decision trees, and random forests, as highlighted in (Trivedi 2020).

When evaluating the performance across two additional datasets, our proposed approach shows similarity to state-of-the-art methods in terms of the Percentage of Correctly Classified Cases and Brier Score, ranking within the mid-range. This indicates that the dimension-reduction-aided approach yields results comparable to those of the prediction-centric approaches in credit scoring. Although comparisons have been made with existing methods, it is crucial to emphasize that the fundamental goal of our approach differs significantly from that of the others. Our primary focus was on integrating dimension reduction techniques to address the challenges posed by a large number of predictors, setting our approach apart. Therefore, such comparisons should be approached with caution considering the distinct objectives and methodologies employed in the evaluation.

In our decision-making process, which relies on a single variable, it is important to clarify that this choice is guided by specific criteria tailored for our dimension-reduction

**Table 13** Comparison to state-of-the-art approaches for HMEQ data via AUC, PCC, and BS

| Methods | AUC | PCC | BS |
|---|---|---|---|
| MD-SDR ($\hat{d}_{mmrn} = 1$) | 0.71 | 0.81 | 0.12 |
| PCA($\hat{d}_{pca} = 1$) | 0.55 | 0.71 | 0.16 |
| PCA ($\hat{d}_{pca} = 14$) | 0.70 | 0.80 | 0.12 |
| Linear logistic regression | 0.79 | 0.81 | 0.12 |
| Non-linear logistic regression | 0.80 | 0.82 | 0.12 |
| Non-linear logistic regression + ALasso | 0.81 | 0.82 | 0.11 |
| Random forest | 0.94 | 0.90 | 0.07 |
| PLTR | 0.90 | 0.88 | 0.08 |
| Support vector machine | 0.80 | 0.81 | 0.13 |
| Neural network | 0.79 | 0.81 | 0.12 |

**Table 14** Comparison to state-of-the-art approaches for ACA data via AUC, PCC, and BS

| Methods | AUC | PCC | BS |
|---|---|---|---|
| MD-SDR ($\hat{d}_{mmrn} = 1$) | 0.85 | 0.87 | 0.11 |
| PCA($\hat{d}_{pca} = 1$) | 0.80 | 0.80 | 0.14 |
| PCA ($\hat{d}_{pca} = 22$) | 0.86 | 0.86 | 0.10 |
| Linear logistic regression | 0.90 | 0.84 | 0.12 |
| Non-linear logistic regression | 0.61 | 0.61 | 0.39 |
| Non-linear logistic regression + ALasso | 0.89 | 0.82 | 0.13 |
| Random forest | 0.93 | 0.86 | 0.10 |
| PLTR | 0.93 | 0.86 | 0.10 |
| Support vector machine | 0.92 | 0.84 | 0.11 |
| Neural network | 0.91 | 0.85 | 0.11 |

approach, particularly the subspace distance for the DCOV-based SDR. This criterion ensures the preservation of the crucial regression information. In this study, the decision to grant a loan to a consumer is framed as a classification problem using logistic regression. During the dimension reduction procedure, essential information for decision making is consolidated into a single reduced variable (a linear combination of all original predictors), considering the logistic regression related to loan approval. For banks, using the observed predictor values for a new customer generates an updated reduced variable, enabling the prediction of the outcome through logistic regression for that customer.

Future research may involve exploring the computation of credit scores for individual customers by applying various dimension-reduction methods. The average credit score is determined by assigning weights to each method employed to calculate the final credit score for a customer. This approach avoids the reliance on a single-variable selection method, as discussed in Miljkovic and Grün (2021). By incorporating the uncertainty associated with different methods, model averaging allows for the modeling of the effect

of methodological uncertainty in the credit scoring process. Another potential avenue for future research is the development of scorecards specifically designed for continuous factors. Although the categorized variables provide interpretability advantages, the process of discretization may result in some loss of information.

## Appendix

This section presents a comparison between our method and the state-of-the-art approaches outlined in Dumitrescu et al. (2022). While most efforts have been made to ensure a fair comparison, it is important for readers to note that the datasets used in this study may differ slightly from those used in Dumitrescu et al. (2022). Nevertheless, any resulting discrepancies were anticipated to be negligible. Similar to Dumitrescu et al. (2022), instead of employing ten-fold cross-validation, $N \times 2$-fold cross-validation with $N = 5$ was adopted to compute criteria such as the AUC, percentage of correctly classified cases (PCC), and Brier score (BS). This adjustment allows a direct comparison between our approach and the findings of Dumitrescu et al. (2022). Please refer to Dumitrescu et al. (2022) for a detailed introduction to these criteria. All the results for the state-of-the-art methods were directly copied from Dumitrescu et al. (2022). The comparative results of all methods are summarized in Tables 13 and 14 for the HMEQ and ACA datasets. Our proposed approach demonstrates a similarity to state-of-the-art methods in terms of PCC and BS, ranking within the mid-range. Consequently, it can be inferred that the dimension-reduction-aided approach yields results comparable to those of the prediction-centric approaches in credit scoring. Although comparisons were made between the proposed approach and state-of-the-art methods, it is crucial to note that the fundamental goal of our proposed approach differs significantly from the others. Our primary focus lies in integrating dimension-reduction techniques to address the challenges posed by a large number of predictors, separating our approach from the others. Therefore, such comparisons should be approached with caution, and the different objectives and methodologies employed in the evaluation should be carefully considered. Note that the criterion AUC is calculated for our approach and PCA using functions in the R package `pROC`, developed by Xavier et al. (2023).

**Abbreviations**

| | |
|---|---|
| DRA-CS | Dimension reduction-assisted credit scoring |
| SMEs | Small and medium-sized enterprises |
| PCA | Principal component analysis |
| DCOV-SDR | Sufficient distance-based dimension reduction |
| MMRN | Riemannian-Newton optimization algorithm with the Majorization-Minimization method |
| MD-SDR | DCOV-SDR through MMRN |
| SVM | Support vector machines |
| SDR | Sufficient dimensionality reduction |
| DCOV | Distance covariance |
| MM | Majorization-Minimization |
| CPVE | Cumulative percentage variance |
| PCs | Principal component |
| SW | Stepwise regression |
| The PA | Prediction accuracy |
| EM | Expectation maximization |
| DNN | Deep neural network |
| SQP | Sequential quadratic programming |

**Acknowledgements**
We would like to thank Dr. Jens Mueller of Miami University's Research Computing Support team for assisting with the computations done in this project on the Redhawk cluster. We appreciate helpful comments received from Ms. Lisa Werwinski and Dr. Robert Davis while editing and proofreading this paper. Many thanks to Dr. Jacques Rioux from the SAS Institute Inc. for providing the HMEQ data to us. We are also grateful for the valuable suggestions and comments from the editors and reviewers.

**Author Contributions**
The authors share equal contribution and both authors read and approved the final manuscript.

**Funding**
Not applicable.

**Availability of data and materials**
The analysis in R is available per readers' request.

## Declarations

**Competing interests**
The authors declare that they have no Conflict of interest.

Received: 17 May 2023 Accepted: 11 November 2024
Published online: 07 January 2025

**References**

Chen X, Sheng W, Yin X (2018) Efficient sparse estimate of sufficient dimension reduction in high dimension. Technometrics 60(2):161–168
Cook RD (1998) Regression graphics: ideas for studying regressions through graphics, vol 318. John Wiley & Sons, NJ
Cook RD, Weisberg S (1991) Discussion of sliced inverse regression for dimension reduction. J Am Stat Assoc 86(414):328–332
Deng X, Wang S (2023) A two-stage framework for credit scoring based on feature augmentation and dimension reduction. Commun Stat Simul Comput. https://doi.org/10.1080/03610918.2023.2234677
Duan N, Li KC (1991) Slicing regression: a link-free regression method. Ann Stat 19:505–530
Dumitrescu E, Hué S, Hurlin C, Tokpavi S (2022) Machine learning for credit scoring: improving logistic regression with non-linear decision-tree effects. Euro J Oper Res 297(3):1178–1192
Dutang C, Charpentier A (2016) CASdatasets. http://cas.uqam.ca/pub/R/, R package version 1.0-6
Emekter R, Tu Y, Jirasakuldech B, Lu M (2015) Evaluating credit risk and loan performance in online peer-to-peer (P2P) lending. Appl Econ 47(1):54–70
Giannouli P, Karagrigoriou A, Kountzakis CE, Ntotsis K (2021) Multilevel dimension reduction for credit scoring modelling and prediction: empirical evidence for Greece. Commun Stat: Case Stud Data Anal Appl 7(4):545–560
Guiso L, Sapienza P, Zingales L (2013) The determinants of attitudes toward strategic default on mortgages. J Finance 68(4):1473–1515
Gunnarsson BR, Vanden Broucke S, Baesens B, Óskarsdóttir M, Lemahieu W (2021) Deep learning for credit scoring: do or don't? Euro J Oper Res 295(1):292–305
Hand DJ, Henley WE (1997) Statistical classification methods in consumer credit scoring: a review. J R Stat Soc: Ser A (Stat Soc) 160(3):523–541
Khandani AE, Kim AJ, Lo AW (2010) Consumer credit-risk models via machine-learning algorithms. J Bank Finance 34(11):2767–2787
King RD, Feng C, Sutherland A (1995) Statlog: comparison of classification algorithms on large real-world problems. Appl Artif Intell Int J 9(3):289–333
Laborda J, Ryoo S (2021) Feature selection in a credit scoring model. Mathematics 9(7):746
Li KC (1991) Sliced inverse regression for dimension reduction. J Am Stat Assoc 86(414):316–327
Liu Y, Schumann M (2005) Data mining feature selection for credit scoring models. J Oper Res Soc 56(9):1099–1108
Markov A, Seleznyova Z, Lapshin V (2022) Credit scoring methods: latest trends and points to consider. J Finance Data Sci 8:180–201
Matignon R (2007) Data mining using SAS enterprise miner. John Wiley & Sons, NJ
Miljkovic T, Grün B (2021) Using model averaging to determine suitable risk measure estimates. North Am Act J 25(4):562–579
Ogundimu EO (2022) On lasso and adaptive lasso for non-random sample in credit scoring. Stat Model 24:147–181
Ortega JM, Rheinboldt WC (2000) Iterative solution of nonlinear equations in several variables. SIAM
Pearson K (1901) LIII. On lines and planes of closest fit to systems of points in space. London, Edinburgh, and Dublin Phil Magaz J Sci 2(11):559–572
R Core Team (2022) R: a language and environment for statistical computing. R Foundation for statistical computing, Vienna, Austria, https://www.R-project.org/
Roy PK, Shaw K (2021) A multicriteria credit scoring model for SMEs using hybrid BWM and TOPSIS. Financ Innov 7:1–27
Sadatrasoul S, Gholamian M, Shahanaghi K (2015) Combination of feature selection and optimized fuzzy Apriori rules: the case of credit scoring. Int Arab J Info Technol (IAJIT) 12(2):138–145

Sheng W, Yin X (2013) Direction estimation in single-index models via distance covariance. J Multivar Anal 122:148–161
Sheng W, Yin X (2016) Sufficient dimension reduction via distance covariance. J Comput Gr Stat 25(1):91–104
Sun Y, Babu P, Palomar DP (2016) Majorization-minimization algorithms in signal processing, communications, and machine learning. IEEE Trans Signal Process 65(3):794–816
Székely GJ, Rizzo ML, Bakirov NK (2007) Measuring and testing dependence by correlation of distances. Ann Stat 35(6):2769–2794
Tibshirani R (1996) Regression shrinkage and selection via the lasso. J R Stat Soc Ser B: Stat Method 58(1):267–288
Trivedi SK (2020) A study on credit scoring modeling with different feature selection and machine learning approaches. Technol Soc 63:101413
Tufféry S (2011) Data mining and statistics for decision making. John Wiley & Sons, NJ
Ulfarsson MO, Solo V (2014) Selecting the number of principal components with sure. IEEE Signal Process Lett 22(2):239–243
Uyeda JC, Caetano DS, Pennell MW (2015) Comparative analysis of principal components can be misleading. Syst Biol 64(4):677–689
Valle S, Li W, Qin SJ (1999) Selection of the number of principal components: the variance of the reconstruction error criterion with a comparison to other methods. Industr Eng Chem Res 38(11):4389–4401
Viaene S, Dedene G (2005) Cost-sensitive learning and decision making revisited. Euro J Oper Res 166(1):212–220
Wang Q, Xue Y (2023) A structured covariance ensemble for sufficient dimension reduction. Adv Data Anal Class 17(3):777–800
Wang Q, Yin X (2008) A nonlinear multi-dimensional variable selection method for high dimensional data: sparse mave. Comput Stat Data Anal 52(9):4512–4520
Wang J, Hedar AR, Wang S, Ma J (2012) Rough set and scatter search metaheuristic based feature selection for credit scoring. Exp Syst Appl 39(6):6123–6128
Wang P, Yin X, Yuan Q, Kryscio R (2021) Feature filter for estimating central mean subspace and its sparse solution. Comput Stat Data Anal 163:107285
Woo H, Sohn SY (2022) A credit scoring model based on the Myers-Briggs type indicator in online peer-to-peer lending. Financ Innov 8(1):1–19
Wu R, Chen X (2021) MM algorithms for distance covariance based sufficient dimension reduction and sufficient variable selection. Comput Stat Data Anal 155:107089
Xavier R, Natacha T, Alexandre H, Natalia T, Frédérique L, Jean-Charles S, Markus M, Stefan S, Matthias D, Zane B (2023) pROC: display and analyze ROC curves. https://xrobin.github.io/pROC/, R package version 1.18.5
Ye Z, Weiss RE (2003) Using the bootstrap to select one of a new class of dimension reduction methods. J Am Stat Assoc 98(464):968–979
Zou H (2006) The adaptive lasso and its oracle properties. J Am Stat Assoc 101:1418–1429
Zou H, Hastie T (2005) Regularization and variable selection via the elastic net. J R Stat Soc Ser B: Stat Method 67(2):301–320

## Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.